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i. Sieve tube conducts **sucrose** only. But the *photosynthate* in chloroplast mostly in the form of starch or trios-phosphate which has to be transported to the cytoplasm where it will be converted into sucrose for further translocation.

ii. Sucrose moves from mesophyll to nearby sieve elements by short distance transport.

iii. From sieve tube to sink by long-distance transport.

11.7.6 Phloem Unloading

From sieve elements sucrose is translocated into sink organs such as roots, tubers, flowers and fruits and this process is termed as **phloem unloading**. It consists of three steps:

1. ***Sieve element unloading***: Sucrose leave from sieve elements.
2. ***Short distance transport***: Movement of sucrose to sink cells.
3. ***Storage and metabolism***: The final step when sugars are stored or metabolized in sink cells.

11.7.7 Mechanism of Translocation

Several hypotheses have been proposed to explain the mechanism of translocation. Some of them are given below:

1. Diffusion hypothesis

As in diffusion process, this theory states the translocation of food from higher concentration (from the place of synthesis) to lower concentration (to the place of utilization) by the simple physical process. However, the theory was rejected because the speed of translocation is much higher than simple diffusion and translocation is a biological process which any poison can halt.

2. Activated diffusion theory

This theory was first proposed by **Mason** and **Maskell** (1936). According to this theory, the diffusion in sieve tube is accelerated either by activating the diffusing molecules or by reducing the protoplasmic resistance to their diffusion.

3. Electro-Osmotic theory

The theory of electro osmosis was proposed by **Fenson** (1957) and **Spanner** (1958). According to this, an electric-potential across the sieve plate causes the movement of water along with solutes. This theory fails to explain several problems concerning translocation.

4. Munch Mass Flow hypothesis

Mass flow theory was first proposed by **Munch** (1930) and elaborated by **Crafts** (1938). According to this hypothesis, organic substances or solutes move from the region of high osmotic pressure (from mesophyll) to the region of low osmotic pressure along the turgor pressure gradient. The principle involved in this hypothesis can be explained by a simple physical system as shown in figure 11.22.

Two chambers "A" and "B" made up of semipermeable membranes are connected by tube "T" immersed in a reservoir of water. Chamber "A" contains highly concentrated sugar solution while chamber "B" contains dilute sugar solution. The following changes were observed in the system,

i. The high concentration sugar solution of chamber "A" is in a hypertonic state which draws water from the reservoir by endosmosis.

ii. Due to the continuous entry of water into chamber "A", turgor pressure is increased.